

Adversarial Profiled Side-Channel Attack with Unsupervised Domain Adaptation

Fanfei Meng

School of Cyber Science and Engineering
Zhengzhou University
Zhengzhou, China
mengfanfei@gs.zzu.edu.cn

Zhanbo Li

School of Cyber Science and Engineering
Zhengzhou University
Zhengzhou, China
iezbli@zzu.edu.cn

Baolei Mao*

School of Cyber Science and Engineering
Zhengzhou University
Zhengzhou, China
maobaolei@zzu.edu.cn

Wei Hu

School of Cybersecurity
Northwestern Polytechnical University
Xi'an, China
weihu@nwpu.edu.cn

Maoyuan Qin

School of Computer Science and Engineering
Xi'an Technological University
Xi'an, China
qinmaoyuan@xatu.edu.cn

Qiang Zhou

School of Cyber Science and Engineering
Zhengzhou University
Zhengzhou, China
zhouqiang@gs.zzu.edu.cn

Abstract—Deep learning-based methods have been proven to be very successful when applied to profiled side-channel attacks (SCA). In profiled SCA, the attacker gains knowledge about the target device by accessing a profiling device before the attack. However, most of the work does not consider the problem that it is difficult to obtain profiling traces that is fully consistent with the attack traces because the target device often employs more advanced masking countermeasures. Even if a small discrepancy between profiling traces and attack traces (regarded as domain discrepancy) may cause the attack failure. To address the problem of domain discrepancy, we propose a side-channel attack method based on the domain adversarial self-attention mechanism (DASA-SCA). Specifically, the method feeds the feature extractor through the bounded loss of the domain classifier when building the model in order to learn a hidden representation that is not only discriminative but also domain invariant. To enhance this domain-invariant representation, we introduce self-attention mechanism in the feature extractor to give higher weights to the significant parts of the leakage in order to improve the feature extraction. We implement three different deep learning models including CNN, ResNet, and LSTM in the DASA-SCA approach for both electromagnetic(EM) side-channel and power side-channel to demonstrate the scalability and generality of the framework. Finally, we validate the effectiveness of the DASA-SCA approach using the ASCAD dataset and DPA dataset. The experimental results demonstrate that our method can significantly reduce the impact of domain discrepancy caused by different countermeasures and enhance the side-channel attack capability.

Index Terms—Side-channel Attacks, Domain Adversarial, Deep Learning, Self-Attention, ASCAD

I. INTRODUCTION

Side-channel attacks pose significant threats to cryptographic devices, attracting increasing attention from academia

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*Corresponding author

and industry [1]. Among various types of side-channel attacks, the profiled attacks have proven to be highly effective. Such attacks assume that the attacker possesses a profiling device identical to the target device and can collect the traces to construct an attack model. This model is subsequently utilized to extract sensitive information from the target device by analyzing the side-channel information [2]. Deep learning methods have shown great potential in enhancing the effectiveness of side-channel attacks. For instance, Maghrebi et al. pioneered Deep Learning-based Side-channel Attacks (DL-SCA) in 2016 and demonstrated its effectiveness and significant potential [3]. Deep learning facilitates the processing of high-dimensional inputs and partly mitigates the impact of cryptographic algorithm masking strategies on key recovery.

Note that a prerequisite for a profiled attack to be successful is that the profiling device and the target device are as similar as possible. In many research works and publicly available SCA datasets, the profiling traces and attack traces come from the same device. However, in realistic SCA scenario, the attack traces must be measured from the target device. Although the cryptographic algorithms running on the profiling device and the target device are the same, their algorithmic implementations may be different. For example, turning on or off the countermeasures on the target device inevitably leads to two different leakage distributions, in which case the deep learning model trained on the profiling device does not generalize well to the target device.

Transfer learning has emerged as a promising approach to tackle this problem. Recent studies have shown that transfer learning can be used in deep learning-based side-channel attacks to achieve better performance than traditional approaches [4]. For instance, Yu et al. proposed a meta-transfer learning method that can transfer knowledge from the source domain to the target domain to improve the side-channel attacks [5]. However, their focus is on reducing the number of profiling traces. Moreover, they still use labeled data obtained from the target domain, which is not a realistic profiled SCA

scenario from the attacker's perspective. Cao et al. proposed a method for side-channel attack using domain adaptation but did not pay attention to the timing relationship between the leakage [6].

In conclusion, in practical side-channel attacks, it is often difficult to obtain labeled traces from the target device, this is because direct access to the device or observation of its inputs and outputs is usually restricted due to confidentiality requirements. Moreover, the profiling traces and attack traces can exhibit significant differences in the distribution of critical information leakage, which can be attributed to various countermeasures such as desynchronization, Gaussian Noise, and shuffle. Desynchronization means that each trace is randomly shifted to the left by a random number, Gaussian Noise means introducing a normal-distributed random value to each point of the trace, and shuffle means rearranging the order of the key.

To address these issues, we propose a side-channel attack framework based on the domain adversarial self-attention mechanism (DASA-SCA). Specifically, our approach utilizes CNN, ResNet, and LSTM-based feature extraction networks to extract discriminative features from the traces. We further improve the effectiveness of feature extraction by adding a self-attention mechanism to the feature extractor. Subsequently, we learn domain-invariant features by jointly optimizing the feature extractor and the domain classifier. Finally, we use a label classifier to predict the encryption key based on the features.

To summarize, we make the following contributions:

- We propose a novel approach in profiled side-channel attacks by integrating self-attention mechanism with adversarial domain adaptation (DASA-SCA). This method combines the power of self-attention mechanism, which effectively captures the temporal characteristics of leakage, with the adversarial domain adaptation technique to enhance the feature extraction capability. Notably, this approach improves the attack on the target device without the need for the labeled attack traces.
- We employ three deep neural network models including CNN, ResNet, and LSTM in our DASA-SCA method to demonstrate the generality of our DASA-SCA method.
- We evaluate our method in a scenario of domain differences due to three different countermeasures, including desynchronization, Gaussian noise, and shuffle. Experimental results demonstrate that the DASA-SCA method effectively improves the attack, successfully addressing the challenges posed by various countermeasures.

II. THREAD MODEL

We consider hardware components where the attacker aims to gather critical information through the EM or power channel. Our threat model assumes that the attacker has knowledge of the encryption algorithm running on the target device, and runs the same encryption algorithm on the profiling device. The attacker then obtains a well-performing deep learning model trained on the profiling device, which corresponds

to possessing prior information about the profiling device. We assume that it is difficult to obtain the labeled traces from the target device due to the confidentiality requirement. The adversary has no direct access to the target device and cannot observe the inputs and the outputs. We assume that the distribution of the traces collected from the profiling device and the target device is not consistent. The target device use more advanced countermeasures such as desynchronization, Gaussian noise, and shuffle.

III. PRELIMINARIES

A. Profiled Side-Channel Attacks

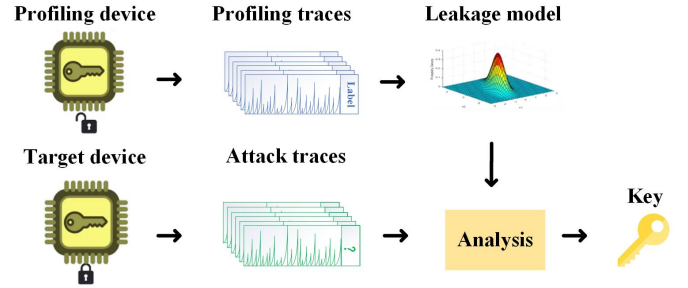


Fig. 1: Profiled Side-Channel Attacks

The profiled attack generally is composed of two phases: a profiling phase and an attack phase as shown in Fig. 1. In the profiling phase, the attacker uses the profiling traces obtained from the profiling device to build the side-channel leakage model, and each trace is labeled because the profiling device is fully controlled by the attacker. In the attack phase, the attacker obtains the attack traces from the target device. The real label of the attack traces is predicted using the leakage model, which is used to achieve the attack purpose of recovering the key.

B. Domain adversarial training

Domain Adversarial Neural Network (DANN) addresses domain adaptation challenges by training a feature extractor and a domain classifier simultaneously. DANN aims to learn domain-invariant representations by introducing the domain adversarial training procedure. The network is encouraged to generate indistinguishable representations between the source domain and the target domain, promoting the feature extractor to acquire domain invariant features. To achieve this, DANN incorporates the domain adversarial loss function. The feature extractor try to minimize the domain adversarial loss when the domain classifier predicts domain labels. This approach facilitates the extraction of task-specific features while disregarding domain-specific variations.

C. Self-Attention mechanism

The attention mechanism allows the model to focus on different parts of the input sequence when computing the output, with the aim of identifying the most relevant parts of the input. The self-attention mechanism extends this concept by allowing the model to focus on different parts of the

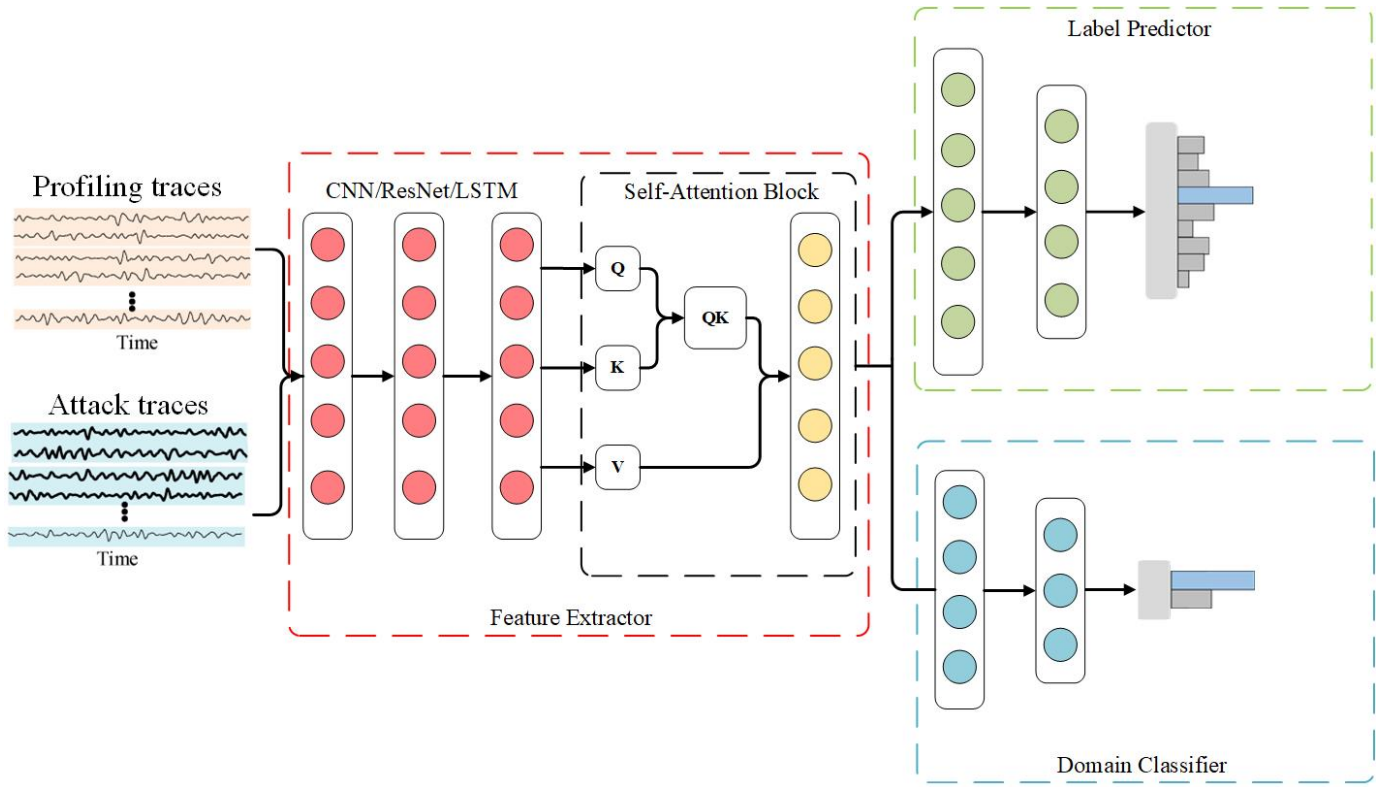


Fig. 2: DASA-SCA framework

same input sequence in order to improve the quality of the learned representations. In side-channel attacks, a large amount of time-series data from the device needs to be processed, and traditional deep learning models may not be able to handle this long sequence data efficiently. In contrast, self-attention-based models can easily process these data because they can adaptively focus on the important information in the sequences. By incorporating self-attention mechanisms into the feature extractors of our proposed DASA-SCA models, the models can learn to selectively focus on the most important parts of the side channel traces, which can improve the key recovery tasks.

IV. OUR FRAMEWORK

Our proposed DASA-SCA framework is shown in Fig. 2, DASA-SCA consists of four main components: the feature extractor, the self-attention block, the label predictor, and the domain classifier. The feature extractor is a conventional deep neural network that is used to extract features from the input data. The self-attention block improves the deep learning-based side-channel attack models and helps the attack models to better capture the important information in the input data, improving the attack model accuracy of predicting keys. The label predictor is used to receive the important features extracted from the feature extractor and performs the key classification task. The domain classifier is a binary classifier which is used to determine whether the input data comes from

the source or the target domain. Each of these four components is described below.

A. Feature Extractor

The feature extractor serves to extract the most representative features from the side-channel information as the kernel of our DASA-SCA framework, which characterizes the behavior of the cryptographic algorithm. These features may contain electromagnetic (EM) side-channel and power side-channel. During the feature extraction process, the DASA-SCA model undergoes training to learn how to extract features from traces and encode them as vector or matrix representations. Subsequently, these extracted features establish a correspondence with the key intermediate values, forming the foundation for the DASA-SCA model's classification accuracy.

We employ three deep learning networks in the DASA-SCA model such as CNN, ResNet, and LSTM. And we evaluate their different effects on different side-channel attacks. Specifically, In the CNN, 5 convolutional layers are used as feature extractors, and all the convolutional layers have a filter size of 11. The Resnet consists of three residual blocks, and the stride for the three residual blocks is [1,2,2]. The LSTM consists of two LSTM layers, both with num_layers of 1.

B. Self-Attention Block

We integrate the self-attention mechanism into our model to perform the side-channel attack. The superiority of this approach lies in its ability to selectively attend to relevant

TABLE I: Profiling side channel dataset (EM and power)

Dataset	Platform	Traces(Features)	Signal Type	Keys	Implementation	Countermeasures
ASCAD_fixkey	ATMeta8515	60000(700)	EM	fixed key	AES-128	1st-order Boolean mask
ASCAD_variablekey	ATMeta8515	300000(1400)	EM	random key	AES-128	1st-order Boolean mask
DPA Contest v4.1	Atmel ATmega-163	100000(435000)	Power	16 different keys	AES-256	RSM
DPA Contest v4.2	Atmel ATmega-163	80000(1704402)	Power	16 different keys	AES-128	RSM+Shuffle

features and suppress irrelevant ones within the same input sequences, which can effectively capture the intricate relationship between different parts of the trace. This representation can not only extract the effective features of the input data but also be used to analyze the internal execution details of cryptographic algorithms in side-channel attacks. Compared to the traditional side-channel attack, the self-attention-based side-channel attack method can more accurately locate the sensitive operations and better-fit traces.

C. Label Predictor

The label predictor is used to extract the sensitive information from the encrypted devices by utilizing side-channel information and recovering the secret keys. Specifically, the label predictor is used to predict the third byte of the AES algorithm. As a result, the label value ranges from 0 to 255 for the 8-bit key byte. A standard multi-layer perceptron (MLP) neural network classifier is employed to predict the label in our framework. The activation function for the input and hidden layers are Selu, and that for the output layer is softmax.

D. Domain Classifier

The domain classifier distinguishes whether the data comes from the source or the target domain and computes the domain classification loss value. Finally, we feed the domain loss value to the feature extractor through the gradient inversion layer so that the feature extractor learns the domain invariant features.

As described in Section III, the domain classification loss function is one of the most important functions in DASA-SCA. It is used to restrict the feature extractor to learn features with domain-invariant properties in the profiling traces and the attack traces. It is achieved by training a domain classifier. The domain adaptive loss function is shown in Equation (1) as follows.

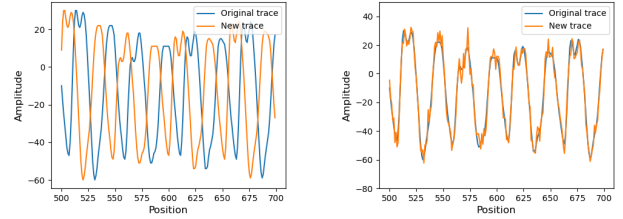
$$L_{domain} = -\frac{1}{M} \sum_{i=1}^M [d_i \log D(\hat{s}_i) + (1 - d_i) \log(1 - D(\hat{s}_i))] \quad (1)$$

where M denotes the total number of data, d_i denotes the domain to which the sample x_i belongs, and $D(\hat{s}_i)$ denotes the prediction of the domain classifier for the feature $\hat{s}_i$.

In the attack framework, we use this loss function to train the feature extractor and the domain classifier as an adversarial model. To achieve domain adaptation, we make the features generated by the feature extractor less sensitive to domain information by minimizing the loss value in the domain classifier.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup



(a) desynchronization

(b) Gaussian noise

Fig. 3: ASCAD with different countermeasures

We use four different datasets in our experiments as shown in Table I. The Advanced Encryption Standard (AES) [7] is used in all the four datasets, but their implementations are different. We conduct different experiments where the target device applies desynchronization (see Fig.3a), Gaussian noise (see Fig.3b) and Shuffle countermeasures, respectively.

ASCAD is an open database used for deep learning techniques evaluation in side-channel attack [8]. It consists of a fixed-key dataset and a variable-key dataset. Each dataset contains three sub-datasets: in ASCAD, the traces are synchronized, ASCAD_desync50.h5 contains the traces with 50 samples window maximum jitter, and ASCAD_desync100 contains the traces with 100 samples window maximum jitter. The ASCAD dataset can be found at <https://github.com/ANSSIFR/ASCAD>.

The strategy of introducing Gaussian Noise to the ASCAD dataset has been adopted to enhance security [9]. In order to evaluate the efficacy of adding noise, a target domain is constructed by introducing a normal-distributed random value $r \sim N(0, var)$ to each point of the trace. In this paper, we set the values of var to 2 and 4, respectively. This methodology serves to demonstrate the impact of incorporating Gaussian Noise on the ASCAD dataset.

The DPA Contest v4 dataset involves two different AES implementations with 1st-order Boolean mask [10]. It includes two datasets, available at <https://www.dpacontest.org/v4/index.php>. DPA Contest v4.1 contains the power traces from AES-256 on Atmel ATmega-163 smart cards with rotating S-box masking. DPA Contest v4.2 consists of the power traces from AES-128 with both RSM and Shuffling strategies.

B. Evaluation Metrics

The key rank is a crucial metric in side-channel attacks (SCA) for assessing the attack based on the best guess

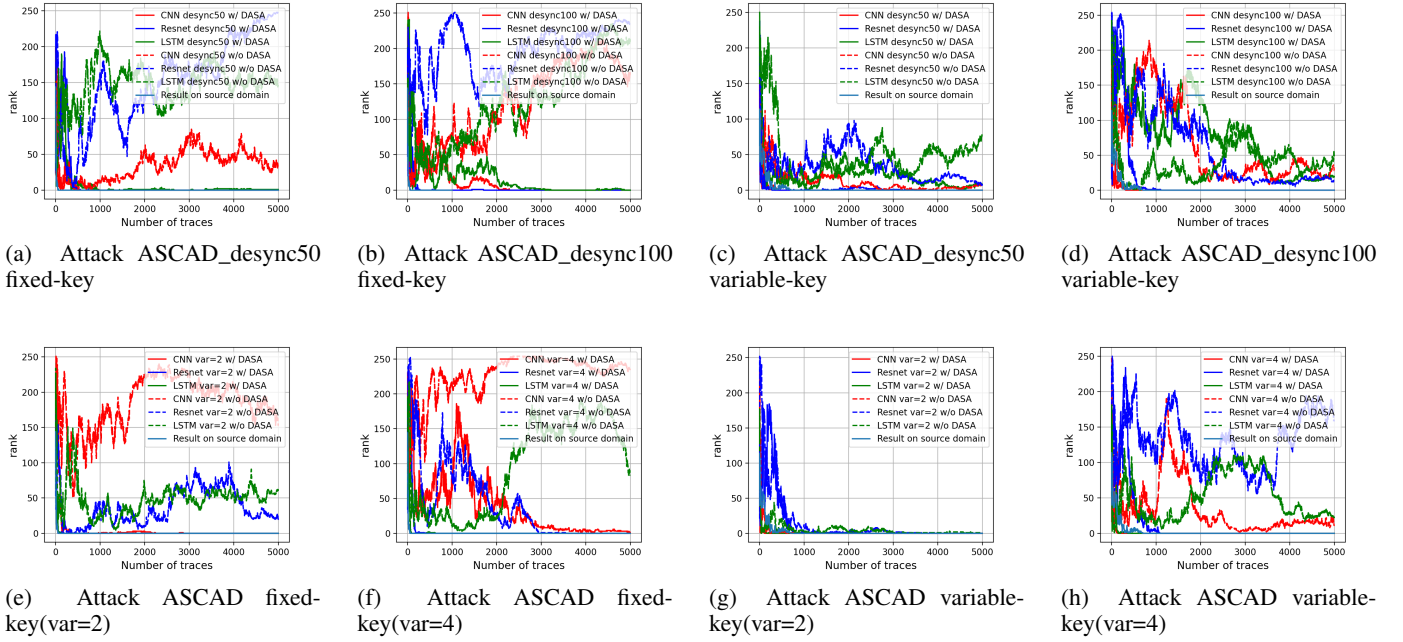


Fig. 4: DASA-SCA attack result on ASCAD with desynchronization(top) or Gaussian noise(bottom)

probability. By analyzing traces, the goal is to infer the real key value in a side-channel attack. This involves treating the key as a variable and estimating it using a neural network.

Using the AES-128 in the ASCAD dataset as an example, we calculate the best guess probability for each hypothetical key byte value. Ranking these probabilities yields the key rank. For each assumed key byte value ranging from 0 to 255, the prediction probability is computed by the neural network which indicates the likelihood of it being the true key:

$$\text{rank}(k) = \sum_{i=0}^{255} [\text{prob}(k)_i \geq \text{prob}(k^*)] \quad (2)$$

Here, $\text{prob}(k)_i$ denotes the best guess probability for hypothetical key i , and $\text{prob}(k^*)$ is the guess probability when the hypothetical key equals the real key. The rank 0 signifies the successful key byte.

C. Attack Results for ASCAD

TABLE II: RESULTS of DASA-SCA ATTACK ASCAD

Dataset	Deep learning Model	desynchronization		Gaussian Noise	
		50	100	2	4
fixed-key	CNN	973	3143	2242	5630
	ResNet	557	1703	175	377
	LSTM	930	3121	194	442
variable-key	CNN	426	636	121	135
	ResNet	1107	3209	1083	1279
	LSTM	7328	*	154	160

¹ *: Unable to recover the correct key

We present the attack results of the ASCAD dataset with desynchronization in Fig.4a, Fig.4b, Fig.4c and Fig.4d. Table II shows the minimum number of traces required for

a successful attack (rank = 0). And the DASA-SCA model use CNN, ResNet, and LSTM as the feature extraction network, respectively. In the fixed-key dataset where the source domain is the ASCAD dataset and the target domain is the ASCAD_desync50 dataset, the DASA-SCA model with ResNet requires 557 traces to recover the correct key byte and the DASA-SCA model with LSTM requires 930 traces to recover the correct key byte. When the target domain is ASCAD_desync100, the DASA-SCA model with ResNet requires 1703 traces to recover the correct key. In the variable-key dataset where the source domain is the ASCAD variable-key dataset and the target domain is the ASCAD_desync50 variable-key dataset, the DASA-SCA model with CNN requires 426 traces to recover the correct key byte. When the target domain is ASCAD_desync100 variable-key dataset, the DASA-SCA model with CNN requires 636 traces to recover the correct key byte. The experimental results show that the DASA-SCA model with ResNet feature extraction network performs best in the ASCAD fixed-key dataset and the DASA-SCA model with CNN feature extraction network performs best in the ASCAD variable-key dataset.

We also present the attack results of the ASCAD dataset with Gaussian Noise in Fig.4e, Fig.4f, Fig.4g and Fig.4h. and Table II. As Fig.4 shown, the target device cannot be successfully attacked without DASA-SCA in most scenarios except for the ASCAD variable-key(var = 2) dataset. However, when we employ proposed DASA-SCA method, the attack result can be improved significantly. When the Gaussian Noise is set to 2, the DASA-SCA model with ResNet requires 175 traces to recover the correct key byte and the DASA-SCA model with LSTM requires 194 traces for the fixed-key

dataset. Moreover, when the Gaussian Noise is set to 4, the DASA-SCA model with ResNet requires 377 traces to recover the correct key byte and the DASA-SCA model with LSTM requires 442 traces for the fixed-key dataset, which are much better than the CNN feature extraction network that requires more than 5000 traces to perform the successful attack. In the variable-key dataset, when the Gaussian Noise is set to 2, DASA-SCA model with CNN requires 121 traces to recover the correct key byte and the DASA-SCA model with LSTM requires 154 traces to recover the correct key byte. When the Gaussian Noise is set to 4, DASA-SCA model with CNN requires 135 traces to recover the correct key byte and the DASA-SCA model with LSTM requires 160 traces to recover the correct key byte.

D. Attack Results for DPA

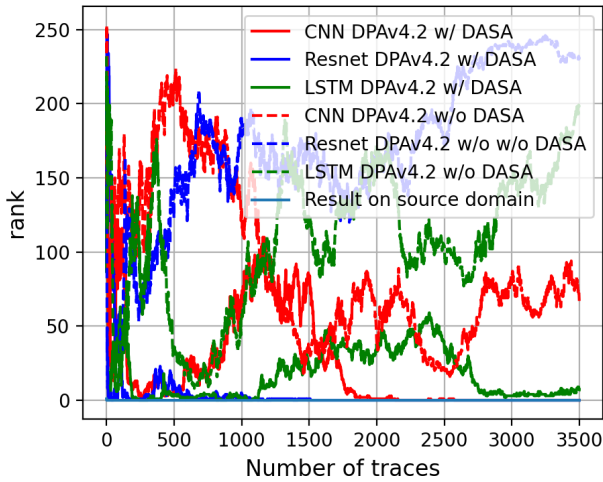


Fig. 5: DASA-SCA attack result on DPA with Shuffle

In this subsection, We use the power traces from DPA Contest v4.1 as the profiling traces and the traces from DPA Contest v4.2 with shuffle countermeasures as the attack traces. The attack results for the DPA contest v4.2 are presented in Fig. 5, the DASA-SCA model with ResNet can successfully attack the power side channel with 1509 traces. And The DASA-SCA model with CNN can perform the successful attack with 2571 traces. The experimental results demonstrate the powerful attack capability of our proposed DASA-SCA framework in power side channel analysis.

VI. CONCLUSION

We focus on addressing the open question of profiled SCA using unsupervised adversarial domain adaptation techniques in this article. We consider the issue of the discrepancy between the profiling traces and the attack traces as domain discrepancy. In order to deal with this domain discrepancy, we propose a new attack framework called DASA-SCA which integrates self-attention mechanism into the adversarial domain adaptation framework. We add a domain classifier that makes

the framework eventually learn a representation that is not only discriminative but also domain-invariant and adopt the domain loss as the penalty of the cross-entropy loss function. we also employ CNN, ResNet, and LSTM as the feature extraction networks in our DASA-SCA framework to evaluate its scalability and generality. We also perform side channel attack on the ASCAD and DPA datasets by our DASA-SCA method and perform the attack on the scenarios of domain differences resulting from desynchronization, Gaussian noise, and Shuffle countermeasures. The experimental results show that DASA-SCA can significantly reduce the amount of traces for efficient SCA attacks and the DASA-SCA framework is scalability and generality for different attack scenarios. In the future, we will focus on exploring more domain adaptation methods and validate the powerful attack capability of domain adaptation in side channel attack techniques.

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